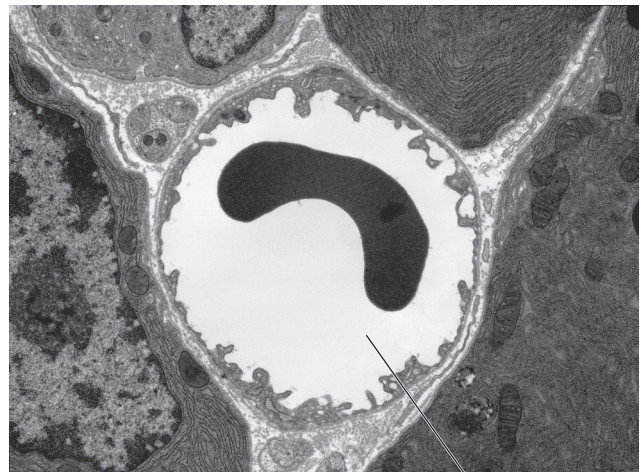


Arterioles branch to form capillaries that supply blood to tissues.

(a) Explain the ways in which the structure of an artery is adapted to its function.

..... [4]

arteriole magnification $\times 2000$



capillary magnification $\times 6500$

(b) (i) Identify the cells inside the lumen of the arteriole in Fig. 4.1 **and** state **one** reason for your identification.

..... [2]

(ii) Describe the differences between the arteriole and the capillary that are **visible** in Fig. 4.1.

[4]

Fig. 4.2 shows a capillary network in a mammalian tissue.

The arrows indicate the direction of flow of body fluids.

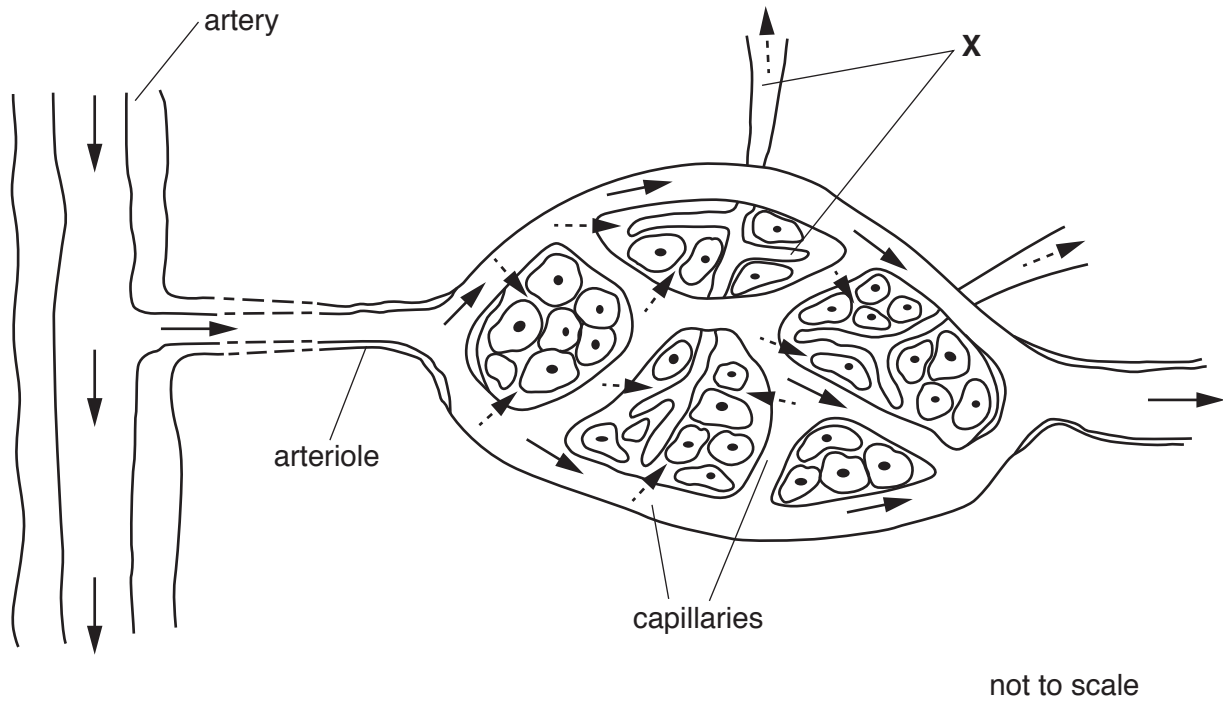


Fig. 4.2

- (c) (i) Capillaries have a role in the formation of tissue fluid.

Explain how tissue fluid is formed in the capillary network.

.....

.....

.....

.....

..... [2]

- (ii) The vessels labelled **X** in Fig. 4.2 carry excess tissue fluid back into the circulatory system.

Name the fluid inside the vessels labelled **X** and state **one** way in which its composition differs from blood plasma.

name of fluid

difference

.....

[2]

[Total: 14]